

Teaching Strategies: Integrating AI Thoughtfully

Practical approaches for incorporating AI into your teaching practice

Start Where You Are

Week 1: Personal Exploration

Before asking students to use AI, try it yourself: - Generate quiz questions for your next topic - Create discussion prompts for tutorials - Draft assignment rubrics - Summarise key readings

Document: What worked? What needed fixing? How long did it take?

Week 2: Transparent Modeling

Show students your process: - “I used AI to generate these practice problems, then I...” - “Here’s the original AI output and here’s why I changed it” - “This rubric started with AI, but needed human expertise for...”

Key: Demonstrate critical evaluation, not blind acceptance

Redesigning Assessments for an AI World

From “What” to “How and Why”

Traditional Assessment: “Write a 2000-word essay on marketing strategy”

AI-Aware Assessment Options:

Option A: Process Documentation - Submit AI conversation transcript - Annotate where AI was wrong/limited - Reflection on how human expertise improved output - Final submission shows evolution of thinking

Option B: In-Class Components - AI-generated draft submitted before class - In-class critique and improvement - Peer review of AI use strategies - Live presentation of key insights

Option C: Personal Application - Connect theory to personal experience - Include specific local context - Require primary data collection - Emphasise unique perspective

Practical Classroom Activities

1. AI Error Hunt (15 minutes)

Setup: Generate course content with deliberate errors **Activity:** Students identify and correct mistakes **Learning:** Critical evaluation, subject mastery **Debrief:** Why did AI make these errors?

2. Prompt Engineering Challenge (20 minutes)

Setup: Same question, groups craft different prompts **Activity:** Compare AI outputs from various prompts **Learning:** Importance of clear communication **Debrief:** What made prompts more/less effective?

3. Human vs AI Debate (30 minutes)

Setup: AI generates one side of argument **Activity:** Students argue opposite position **Learning:** Critical thinking, argument construction **Debrief:** What did AI miss about context/nuance?

4. AI Improvement Workshop (25 minutes)

Setup: AI creates initial solution/answer **Activity:** Groups enhance with human insight **Learning:** Value of domain expertise **Debrief:** What uniquely human elements were added?

Setting Clear Expectations

Sample Syllabus Statement

AI Use in This Course

Artificial Intelligence tools are part of professional practice in [field]. In this course:

- You MAY use AI for brainstorming, drafting, and exploration
- You MUST disclose all AI use in assignments
- You MUST verify all AI-generated content

- You CANNOT submit unedited AI content as your own work

Think of AI as a research assistant—helpful but requiring supervision. Your learning comes from evaluating and improving AI output, not from the output itself.

Disclosure Examples

Simple: “AI assisted with initial research”

Detailed: “Used ChatGPT to generate outline structure, refined with course concepts, verified statistics through library databases”

Comprehensive: Appendix with prompts, outputs, and modification log

Managing Different Scenarios

When Students Over-Rely on AI

Signs: Generic responses, missing personal voice, can’t explain work

Strategies: - Require hand-written components - Add oral examinations - Include reflection on process - Focus on application to specific contexts

When Students Fear Using AI

Signs: Anxious about “cheating”, falling behind peers

Strategies: - Model your own AI use - Start with low-stakes activities - Provide clear guidelines - Share success examples

When AI Gives Wrong Information

Opportunity: Teaching moment about verification

Strategies: - Celebrate error detection - Discuss why AI made mistake - Practice fact-checking methods - Emphasise expertise value

Progressive Integration Model

Foundation Level (Weeks 1-4)

- Instructor demonstrates AI use
- Students observe and discuss
- Focus: Understanding capabilities/limitations

Guided Practice (Weeks 5-8)

- Structured AI activities in class
- Clear parameters for use
- Focus: Developing critical evaluation

Independent Application (Weeks 9-12)

- Students choose when/how to use AI
- Justify their decisions
- Focus: Strategic tool selection

Synthesis (Final Weeks)

- Reflect on AI's role in learning
- Evaluate impact on understanding
- Focus: Meta-cognitive awareness

Quick Implementation Ideas

Monday: Lecture Prep

Use AI to generate example scenarios, verify for accuracy, share process with students

Tuesday: Tutorial Planning

Create discussion questions with AI, add context-specific elements

Wednesday: Assignment Design

Draft rubric with AI, modify for course objectives

Thursday: Student Support

Use AI to create study guides, add personal expertise

Friday: Reflection

Document what worked, adjust for next week

Addressing Common Concerns

“Students will cheat” → Design assessments that require personal insight, process documentation, or live demonstration

“AI undermines learning” → Focus on higher-order thinking: evaluation, synthesis, creation

“I don’t understand AI” → Start small, learn alongside students, share your journey

“It’s not fair if some students use AI” → Make AI use transparent and optional, provide alternatives

Assessment Rubric Adaptation

Traditional Criteria → AI-Era Criteria

Knowledge Display → **Knowledge Application** - Can recite facts → Can verify AI facts - Memorised content → Contextualised understanding

Writing Quality → **Thinking Quality** - Grammar/spelling → Argument construction - Word count → Idea development

Research Skills → **Evaluation Skills** - Found sources → Assessed AI sources - Citation accuracy → Information verification

Originality → **Synthesis** - Unique phrasing → Unique connections - No plagiarism → Acknowledged AI use

Red Flags vs Green Flags

Red Flags (Need Intervention)

- Cannot explain their work
- Inconsistent voice/knowledge level
- Missing assignment-specific requirements
- No personal connection to content

Green Flags (Good Practice)

- Clear about AI's role
 - Shows iteration and improvement
 - Demonstrates critical thinking
 - Adds personal/local context
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Resources for Further Development

Essential Reading

- Your institution's AI policy
- Discipline-specific AI guidelines
- Student feedback on AI use

Try This Week

1. Generate one lecture example with AI
2. Share the process with students
3. Ask students about their AI use
4. Document one success or failure

Build Community

- Share experiences with colleagues
 - Create discipline-specific prompt library
 - Develop shared assessment strategies
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Final Reminders

You don't need to be an AI expert — You need to be a learning expert

Perfect is the enemy of good — Small experiments are valuable

Students are already using AI — Better to guide than ignore

Your expertise matters more, not less — AI makes human judgment crucial

Focus on learning outcomes — Tools change, objectives remain

Resources & References

- (Heaven, W. (2023, April 6). ChatGPT is going to change education, not destroy it. MIT Technology Review. Retrieved September 4 2025, from <https://www.technologyreview.com/2023/04/06/1070216/chatgpt-is-going-to-change-education-not-destroy-it>)[<https://www.technologyreview.com/2023/04/06/1071059/chatgpt-change-not-destroy-education-openai/>]
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 - (Russell Group. (2023). Russell Group principles on the use of generative AI tools in education. Russell Group. Retrieved September 4 2025, from <https://www.russellgroup.ac.uk/policy/policy-briefings/principles-use-generative-ai-tools-education>)[<https://www.russellgroup.ac.uk/policy/policy-briefings/principles-use-generative-ai-tools-education>]
 - [TEQSA on Generative AI](#)
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